🔑 **Answer Key — Teacher Copy · Lesson 2-1**

MSTAR PRACTICE WORKSHEET

Lesson 2-1: Understand Statistical Questions

Grade 6 Mathematics · Standard 6.DS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which of the following is a statistical question?

- **A** How many points did each player score this season?
- **B** How many points are scored for a free throw?
- **C** What sport does this team play?
- **D** How many halves are in a soccer game?

"How many points did each player score?" is statistical because different players will have different scores — there is variability in the data.

Question 2 SELECTED RESPONSE — CORRECT: C

Which question is NOT a statistical question?

- **A** How many home runs did each player hit this year?
- **B** How many miles does each runner train per week?
- **C** How many innings are in a baseball game?
- **D** How tall is each player on the basketball team?

"How many innings are in a baseball game?" has one fixed answer (9) — no variability. The other questions all produce different answers for different people.

Question 3 SELECTED RESPONSE — CORRECT: A

Why is 'How many hours of sleep did each student get last night?' a statistical question?

- (A) Because different students got different amounts of sleep
- (B) Because it asks about sleep
- (C) Because hours are numbers
- (D) Because the answer is always 8 hours

A statistical question anticipates variability — different students slept different amounts, so the data will vary.

Question 4 SELECTED RESPONSE — CORRECT: C

A coach asks: "How many hours does each athlete on the track team sleep the night before a meet?" Why is this a statistical question?

- (A) Because it is about sports
- (B) Because it has a numerical answer
- (C) Because the answers will vary from athlete to athlete
- (D) Because the coach asked it

A statistical question anticipates variability. Different athletes sleep different amounts, so the data collected will vary.

Question 5 SELECTED RESPONSE — CORRECT: D

Which of these is a statistical question?

- (A) What time is it right now?
- (B) What is my shoe size?
- (C) What is 7 × 8?
- (D) How tall are the students in my class?

A statistical question anticipates variability — students' heights differ, so the data will vary.

Question 6 SELECTED RESPONSE — CORRECT: A

Which of these would collect data that VARIES (making it a statistical question)?

A How many siblings does each student in the class have?

B How many days are in one week?

C How many minutes are in one hour?

D What is 12 + 12?

Number of siblings varies from student to student, so the collected data will vary — that makes it statistical. The others each have one fixed answer.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Which question is a STATISTICAL question?

A. How many minutes did I sleep last night?

B. How many students are in our class?

C. What time does school start?

D. How many minutes did each student in our class sleep last night?

A statistical question anticipates answers that VARY. Different students slept different amounts, so the data will vary. The other three each have exactly one fixed answer.

- **A:** This asks about one person, so it has exactly one answer. A statistical question anticipates answers that vary across a group.
- **B:** A class has one size, so this has a single answer. Statistical questions expect the answers to differ.
- **C:** School starts at one fixed time, so there is nothing to vary. Ask something that would give different answers from different students.

Part B — correct answer: C

What makes the other three questions non-statistical?

A. They are about school instead of about students.

B. They use smaller numbers.

C. Each has a single fixed answer, so there is no variability to describe.

D. They cannot be answered with data.

Variability is the whole point. A question with one fixed answer produces one value, and there is no distribution to summarize.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D

Select ALL questions that are statistical questions:

A. How tall is each player on the team?

B. How many hours per week do sixth graders spend reading?

C. What is my shoe size?

D. How long does it take students in our class to get to school?

E. What is the capital of Maryland?

A, B, and D each anticipate answers that vary across people. C has one fixed answer about one person, and E has one fixed factual answer.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Write one statistical question and one non-statistical question about your school's cafeteria. Explain why each question is or is not statistical, and describe what the data would look like for the statistical question.

Model answer: A strong statistical cafeteria question asks about each person, such as 'How many minutes does each student wait in the lunch line?' — the data would be a spread of values like 2, 5, 7, and 11 minutes. A non-statistical one, such as 'How many tables are in the cafeteria?', has a single fixed answer, so there is nothing to collect or compare.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
